Muthoot Institute of Technology & Science, Varikoli
Department of Artificial Intelligence & Data Science
B. Tech. Computer Science and Artificial Intelligence - Semester 7
CAD : Project
ABSTRACT

Marc George : MUT22CA043
Merlin Sarah Jiju : MUT22CA045
Timon K John : MUT22CA068

July 19, 2025

SentiCore: A Neuromorphic AI for Emotion Synthesis and Empathy Modeling using Spiking Neural Networks

Keywords: Spiking Neural Networks, Emotion Simulation, Neuromorphic Software, Multimodal Emotion Perception, Temporal Memory, Empathy Modeling, Surrogate-Gradient Training, Generative Tone-Aware Responses

1 Abstract

Problem Statement: Traditional emotion-aware AI systems merely detect human affect from text, audio, or vision, but they do not internally simulate or evolve emotional states over time. They lack temporal memory, contextual empathy modeling, and generate only reactive outputs, resulting in one-dimensional, non-adaptive interactions. This limitation prevents such systems from maintaining a coherent emotional "persona" that can learn from past interactions and adjust its responses in future conversations.

Proposed Solution: SentiCore is a fully software-based, neuromorphic-inspired framework that encodes multimodal cues—speech tone, facial embeddings, and text sentiment—into spike trains; processes them in a recurrent spiking network to simulate evolving emotions; and conditions a GPT-style text generator for tone-aware, empathetic responses. Synaptic plasticity (surrogate gradients, STDP) enables mood dynamics, while an RL-driven empathy module tunes mirroring versus soothing behaviors. By continuously updating its internal state based on feedback and user cues, SentiCore can demonstrate consistent emotional continuity and deepen its rapport over extended interactions. This approach yields explainable, emotional intelligence suitable for digital companions, mental health assistants, and adaptive tutors.

2 Technologies using

- Python — Core language for SNN modeling, data processing, and API services.
- PyTorch — Deep learning framework powering surrogate-gradient training of spiking networks.
- SpikingJelly — High-performance library for building and training LIF-based SNN layers on GPU.
- openSMILE — Toolkit for extracting audio emotion features from speech.
- Hugging Face Transformers — Pretrained GPT-style models for generative, tone-conditioned text output.
- LangChain — Orchestrates multimodal pipelines, chaining SNN state vectors into LLM prompts.
- BindsNET — Reinforcement-learning library for shaping empathy behaviors via reward feedback.

HOD, AI&DS

References

[1] FeiFan Xu, Deng Pan, Haohao Zheng, Yu Ouyang, Zhe Jia, Hong Zeng, "EESCN: A novel spiking neural network method for EEG-based emotion recognition", *Computer Methods and Programs in Biomedicine*, Volume 243, January 2024

[2] Binqiang Wang, Xiaoqiang Liang, "Incorporating Spiking Neural Network for Dynamic Vision Emotion Analysis", *Lecture Notes in Computer Science*, December 2023

[3] Rebecca Mobbs, Dimitrios Makris, Vasileios Argyriou, "Emotion Recognition and Generation: A Comprehensive Review of Face, Speech, and Text Modalities", *arXiv Preprint*, February 2025

Project Coordinators

Ms. Rinu Rose George & Ms. Thasni KN,
Assistant Professor
Dept. of AI & DS
MITS

Project Guide

Ms. Philo Sumi
Assistant Professor
Dept. of AI & DS
MITS